

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	A	Date:	24-07-2026

Code of Conduct

I V. Bhavana (192311379) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V. Bhavana

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

66
100

Questions

- Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.
- Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.
- Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.
- Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log_{10}$ of the ratio).
- Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.
- Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?
- Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?
- Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?
- Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?
- Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.
- Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?
- Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?
- Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.
- Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.
- Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?
- Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?
- Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?
- Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?
- Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

- Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?
- Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?
- Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?
- Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?
- Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?
- Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?
- Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?
- Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?
- Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?
- Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?
- Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?
- Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?
- Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?
- Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?
- Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?
- Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?
- Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?
- Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?
- Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?
- Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?
- Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km^2 of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, $\text{area} = \pi R^2$) and needs to blanket a 350 km^2 service region. Roughly how many towers will the rollout require?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Faculty In-charge	Course Coordinator	Head of Department
<i>Dr. S. S. S. S. S.</i> Signature: _____ Date: <u>24/7/24</u>	Signature: _____ Date: _____	Signature: _____ Date: _____

Simplex channel : 25 KHz

Spectrum Allocation : 45 MHz
: 45,000 KHz

Full duplex voice channels = $\frac{\text{Spectrum Allocation}}{\text{Simplex channel} \times 2}$

$$= \frac{45,000}{25 \times 2}$$

$$= \frac{45,000}{50}$$

$$= 900$$

Total voice channels : 560

Reuse clusters : 7

Channels per single cell = $\frac{\text{Total voice channels}}{\text{frequency reuse clusters}}$

$$= \frac{560}{7}$$

$$= 80$$

Total no. of carriers : 100.

Per time slots of voice traffic : 8

Total no. of voice channels = $100 \times 8 = 800$

Base station radiates : 80 W. (P_1)

handset transmits : 4 W. (P_2).

$$\text{Power Gap} = 10 \cdot \log \left(\frac{P_1}{P_2} \right)$$

$$= 10 \cdot \log \left(\frac{80}{4} \right)$$

$$= 10 \log 20 = 13 \text{ dB}$$

5. Serving signal = -60 dBm

co-channel interference = -92 dBm.

$$\begin{aligned}\text{Carrier-to-Interference (C/I) ratio} &= -92 - (-60) \\ &= -92 + 60 \\ &= -28 \text{ dBm.}\end{aligned}$$

6. A operator holds spectrum = 42 MHz = 42,000 KHz

Per channel = 300 KHz

Reuse frequency cell pattern = 7.

No. of channels = $\frac{\text{operator holds spectrum}}{\text{per channel}}$ $\therefore$ Reuse frequency

$$= \frac{42000}{300} \therefore 7.$$

$$= \frac{140}{7} = 20.$$

7. Power received = 64 W.

path-loss exponent = 4.

New received power :- $P_r \propto \frac{1}{d}$

$$P_2 = P_1 \left(\frac{d_1}{d_2} \right)^n$$

$$= 64 \left(\frac{1}{2} \right)^4$$

$$= 64 \left(\frac{1}{24} \right)$$

$$= \frac{64}{16} = 4.$$

8. SNR at new edge = $P_2 t$

n (Inverse square root law) = 2.

$$P_2 t = P_1 (t) \left(\frac{d_2}{d_1} \right)^2 \Rightarrow P_2 t = P_1 (t) \cdot \left(\frac{2}{1} \right)^2$$

$$\Rightarrow P_2 t = 4 \cdot P_1 t.$$

cluster size : 4.

Every cell in the cluster : 15 voice channels.

Total channel count = cluster size $\times$ Every cell in cluster

$$= 15 \times 4$$

$$= 60.$$

10. A vehicle travels : $100 \text{ km/h} = 100 \times \frac{5}{18} = 27.77 \text{ m/s}$.

camped on : 9000 MHz carrier.

Doppler shift = $\frac{\text{speed } v \text{ camped on carrier}}{\text{speed of light}}$

$$= \frac{27.77 \times 3 \times 10^8}{3 \times 10^8}$$

$$= 27.77 \times 3 \times 10^3 = 0.0833 = \underline{\underline{83 \text{ Hz}}}$$

11. TDMA : frame lasting : 5 ms.

time slots per frame : 8.

single slot duration = $\frac{\text{frame lasting}}{\text{Time slots per frame}}$

$$= \frac{5}{8} = 0.625 \text{ ms.}$$

12. Spectrum planners : 24 MHz.

carrier spacing : 200 kHz.

carrier frequencies created :
from allocation

$\frac{\text{carrier spacing}}{\text{spectrum bandwidth}}$

$$= \frac{24 \times 10^6}{200 \times 10^3} = 120.$$

12. Handset transmits: 8 W : 8000 mW

Signal experiences of path loss : 33 dB.

Power of base station (dBm) : $10 \cdot \log_{10}(8000 \text{ mW})$

$$= 10 \times 3.903$$

$$= 39.03$$

$\approx 6 \text{ dBm}$ ✓

14. Reference distance : 1 m.

deployment = 1900 MHz.

cell spaced = 2.5 km.

Free-space path loss = $32.44 + 20 \log_{10}(d) + 20 \log_{10}(f)$

$$= 32.44 + 20 \log_{10}(2.5) + 20 \log_{10}(1900)$$

$$= 32.44 + 7.96 + 65.58$$

$$= 105.98 \text{ dB} \approx 106 \text{ dB}$$
 ✓

15. A cluster consists : 6 cells.

Each cell carry calls : 250 simultaneous

Total call-handling capacity : No. of cells ×

Each cell carrying

$$= 6 \times 250$$

$$= 1500$$
 ✓

16. New call requests per second : 4000.

Average calls : 3 minutes. = $3 \times 60 = 180$.

Active calls in the system : 4000×180

$$= 7,20,000$$
 ✓

∴ Active calls : 7,20,000.

10.01 channels : 150.

Each channel : 25 KHz.

$$\begin{aligned}\text{Total bandwidth} &= 150 \times 25 \text{ KHz} \\ &= 3750 \text{ KHz} \\ &= 3.75 \text{ MHz}.\end{aligned}$$

18. Spectrum guard : 40 KHz.

Adjacent frequency bands : 30.

$$\begin{aligned}\text{Total bandwidth} &= \text{guard band size} \times \text{no. of guard bands} \\ &= 40 \times (30 - 1) \\ &= 29 \times 40 = 1160 \text{ KHz} = 1.16 \text{ MHz}\end{aligned}$$

19. Vehicle cruises at : 90 km/h. $\cdot 90 \times \frac{5}{18} = 25 \text{ m/s}$

Radius region : 1.5 km-radius cells. $= 2 \times 1.5 = 3000 \text{ m}.$

Handoffs occur over : $\frac{3600 \text{ s}}{\text{an hour}}$

$$\begin{aligned}&= \frac{\text{Time per cell}}{\frac{3000}{25}} = \frac{3600}{120} \\ &= 30 \text{ seconds}.\end{aligned}$$

20. Network hierarchy : 8 regional centers.

Each oversees : 40 base stations.

Each base : 250 users.

Total user-handling capacity : Regional centers $\times$ base stations $\times$ users

$$= 8 \times 40 \times 250$$

$$= 80,000.$$

21. Blanket length : 400 sq km

Hexagonal cells radius : 400 m. $\frac{400}{1000} = 0.4 \text{ km}$

Cells required to rollout : $\frac{\text{Total regional area}}{\text{cell area}}$

$$= \frac{400}{R^2} = \frac{400}{(0.4)^2} = \frac{400}{0.16} = 2500 \text{ cells.}$$

22. FDMA allots : 30 KHz to each channel.

Total spectrum pool : 6 MHz : 6000 KHz

No. of channels carved out of this allocation : $\frac{6000}{30}$

$$= 200.$$

23. A market study notes : has : 1200 subscribers.

Each place 4 calls per hour.

Average call : 2.5 minutes.

Total offered traffic : $1200 \times 4 \times 2.5$

$$= 12,000 \text{ min/hr}$$

$$= \frac{12000}{60} = 200 \text{ /hr.}$$

24. Band width of 2.5 GHz deployment : 40 MHz

Each user consumes : 400 KHz

Maximum no of users the

band can serve

$$= 100$$

$$\text{Users} = 100.$$

Carrier base stations each cap up users : 600.
Network comprises : 1200 stations.

Total user capacity : 1200×600
 $= 7,20,000$

26. Paging scheme. Sweeps : 4 cells.

Network : 200-cell.

Worst case : Paging rounding needed

$$\begin{aligned} \text{for allocation: } & \frac{200}{4} \\ & = 50. \end{aligned}$$

0.12.

30. ~~Case~~ ~~new~~ channel frequencies : $\frac{\text{Total bandwidth}}{\text{channel bandwidth}}$
% Reuse frequency
 $= \frac{4800}{200} \cdot 1.4 = 60.$

31. no. of channels : $\frac{\text{Total bandwidth}}{\text{Channel spacing.}}$

$$= \frac{3000}{200} = 150.$$

32. Network logs : 2500.

$$\text{N.I.} = 2500 \times 99\%.$$

$$= 2475.$$

34. cluster user capacity : $900 \times 7 = 6300.$

$$27. \text{ Offered traffic} = 40,000 \times 3$$

$$= 1,20,000$$

$$\text{Traffic blocked} = 1,20,000 - 1,00,000$$

$$\text{Fraction blocked} = \frac{20,000}{1,20,000} = 0.17$$

$$28. \text{ Total subscribers} = 6,00,000$$

$$\text{Roaming users} = \frac{15}{100} \times 6,00,000$$

$$= 90,000$$

$$29. \text{ No. of calls / second} = 12,000$$

$$\text{Handling calls} = 9000$$

$$\text{Capacity} = \frac{\text{Peak traffic}}{\text{Processing rate}}$$

$$= \frac{9000}{12000}$$

$$= 0.75 = 75\%$$

$$33. \text{ Power} = 12W = 12,000 \text{ mW}$$

$$\text{Path loss} = 45 \text{ dB}$$

$$\text{Received Power} = 10 \log_{10}(\text{Power}) = 10 \log_{10}(12,000)$$

$$= 10 \times 4.079$$

$$= 40.79$$

$$\text{Received Power} = 40.79 - 45$$

$$= -4.21 \text{ dBm}$$